A FORMAL ALGEBRAIC SPACE SATISFIES THE SHEAF PROPERTY FOR THE FPQC TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens by saying that it is identical to the proof of a proposition in the chapter on properties of algebraic spaces. That sentence is an attribution, not a pointer in place of an argument: the proof is then displayed in full in the counted span, so the card is collected and the sentence is reproduced verbatim.

Editorial note. The statement carries a slogan environment in this chapter; the chapter's own preamble defines slogan as a comment environment, so its contents are discarded at build time and are not part of the delivered mathematical text. The slogan is reproduced in the source span as delivered.

Editorial note. The proof contains two steps the author marks as omitted: a small argument that the chosen etale cover may be taken to contain the image of the given morphism, and a computation that the two restrictions of the descended morphism agree. Both are reproduced verbatim and both are recorded as obligations of the card.

Editorial note. The proof contains a parenthetical addressed to the reader, offering the alternative of replacing the chosen cover by a disjoint union of formal affines so that it is surjective; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 15 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space, and the diagonal being a representable monomorphism. Same chapter, Definition definition-formal-algebraic-space at proof lines 7093-7095 and Lemma lemma-diagonal-formal-algebraic-space at proof lines 7076-7077. Fixes the objects and supplies the representable monomorphism that the uniqueness argument is built on.
- Etale presentations: representability and separatedness of a morphism from a formal affine cover. Same chapter, Lemma lemma-presentation-representable at proof lines 7110-7111 and Lemma lemma-separated-from-separated at proof lines 7111-7112. Lets the base change of the chosen cover be treated as an etale separated morphism of schemes.

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- Etale separated morphisms are ind-quasi-affine, and ind-quasi-affine descent data are effective. More on Morphisms, Lemma more-morphisms-lemma-etale-separated-ind-quasi-affine at proof lines 7134-7136 (with Morphisms, Lemma morphisms-lemma-etale-locally-quasi-finite) and More on Groupoids, Lemma more-groupoids-lemma-ind-quasi-affine at proof lines 7136-7138. Supplies the effectivity of the descent datum built from the cocycle condition, which is what produces a scheme over the base.
- Descent of surjectivity, of etaleness and of being an isomorphism, and fpqc coverings as universal effective epimorphisms. Descent, Lemma descent-lemma-descending-property-isomorphism at proof lines 7095-7097, and Descent, Lemmas descent-lemma-descending-property-surjective and descent-lemma-descending-property-etale at proof lines 7143-7145, and Descent, Lemma descent-lemma-fpqc-universal-effective-epimorphisms at proof lines 7153-7154. Carries the properties back along the covering and descends the morphism to the base.
- The sheaf property for the fpqc topology is equivalent to the equalizer condition, and the corresponding statement for algebraic spaces. Topologies, Lemma topologies-lemma-sheaf-property-fpqc at proof lines 7066-7068, and Properties of Spaces, Proposition spaces-properties-proposition-sheaf-fpqc together with Properties of Spaces, Lemma spaces-properties-lemma-topology-points at proof lines 7061-7064 and 7101-7103. Fixes the reduction to the equalizer condition and supplies the model proof the author follows.

Source reference. Original definition 023V: sources/stacks-chow/tags.txt, source lines 2756–2756. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02VN: sources/stacks-chow/tags.txt, source lines 3774–3774. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L4: sources/stacks-chow/tags.txt, source lines 3395–3395. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KV: sources/stacks-chow/tags.txt, source lines 3386–3386. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 023Q: sources/stacks-chow/tags.txt, source lines 2751–2751. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2667–2667. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2381–2381. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APX: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3744–3744. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APK: sources/stacks-chow/tags.txt, source lines 13478–13478. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AP9: sources/stacks-chow/tags.txt, source lines 13467–13467. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03WS: sources/stacks-chow/tags.txt, source lines 5123–5123. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03BX: sources/stacks-chow/tags.txt, source lines 4364–4364. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 0APL: sources/stacks-chow/tags.txt, source lines 13479–13479. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 022H: sources/stacks-chow/tags.txt, source lines 2706–2706. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be a formal algebraic space over S. Then X satisfies the sheaf property for the fpqc topology.*

Proof. The proof is **identical** to the proof of Properties of Spaces, Proposition 0APL. Since X is a sheaf for the Zariski topology it suffices to show the following. Given a surjective flat morphism of affines $f : T' \to T$ we have: $X(T)$ is the equalizer of the two maps $X(T') \to X(T' \times_T T')$. See Topologies, Lemma 022H.

Let $a, b : T \to X$ be two morphisms such that $a \circ f = b \circ f$. We have to show $a = b$. Consider the fibre product

$$E = X \times_{\Delta_{X/S}, X \times_S X, (a,b)} T.$$

By Lemma 0AIP the morphism $\Delta_{X/S}$ is a representable monomorphism. Hence $E \to T$ is a monomorphism of schemes. Our assumption that $a \circ f = b \circ f$ implies that $T' \to T$ factors (uniquely) through E. Consider the commutative diagram

$$\begin{array}{ccc} T' \times_T E & \longrightarrow & E \\ \downarrow & \nearrow & \downarrow \\ T' & \longrightarrow & T \end{array}$$

Since the projection $T' \times_T E \to T'$ is a monomorphism with a section we conclude it is an isomorphism. Hence we conclude that $E \to T$ is an isomorphism by Descent, Lemma 02L4. This means $a = b$ as desired.

Next, let $c : T' \to X$ be a morphism such that the two compositions $T' \times_T T' \to T' \to X$ are the same. We have to find a morphism $a : T \to X$ whose composition with $T' \to T$ is c. Choose a formal affine scheme U and an étale morphism $U \to X$ such that the image of $|U| \to |X_{red}|$ contains the image of $|c| : |T'| \to |X_{red}|$. This is possible by Definition 0.1, Properties of Spaces, Lemma 03BX, the fact that a finite union of formal affine algebraic spaces is a formal affine algebraic space, and the fact that $|T'|$ is quasi-compact (small argument omitted). The morphism $U \to X$ is representable by schemes (Lemma 0AIG) and separated (Lemma 0APX). Thus

$$V = U \times_{X,c} T' \longrightarrow T'$$

is an étale and separated morphism of schemes. It is also surjective by our choice of $U \to X$ (if you do not want to argue this you can replace U by a disjoint union of formal affine algebraic spaces so that $U \to X$ is surjective everything else still works as well). The fact that $c \circ \mathrm{pr}_0 = c \circ \mathrm{pr}_1$ means that we obtain a descent datum on $V/T'/T$ (Descent, Definition 023V) because

$$\begin{aligned} V \times_{T'} (T' \times_T T') &= U \times_{X, c \circ \mathrm{pr}_0} (T' \times_T T') \\ &= (T' \times_T T') \times_{c \circ \mathrm{pr}_1, X} U \\ &= (T' \times_T T') \times_{T'} V \end{aligned}$$

The morphism $V \to T'$ is ind-quasi-affine by More on Morphisms, Lemma 0AP9 (because étale morphisms are locally quasi-finite, see Morphisms, Lemma 03WS). By More on Groupoids, Lemma 0APK the descent datum is effective. Say $W \to T$ is a morphism such that there is an isomorphism $\alpha : T' \times_T W \to V$ compatible with the given descent datum on V and the canonical descent datum on $T' \times_T W$. Then

$W \to T$ is surjective and étale (Descent, Lemmas 02KV and 02VN). Consider the composition

$$b' : T' \times_T W \longrightarrow V = U \times_{X,c} T' \longrightarrow U$$

The two compositions $b' \circ (\mathrm{pr}_0, 1), b' \circ (\mathrm{pr}_1, 1) : (T' \times_T T') \times_T W \to T' \times_T W \to U$ agree by our choice of α and the corresponding property of c (computation omitted). Hence b' descends to a morphism $b : W \to U$ by Descent, Lemma 023Q. The diagram

$$\begin{array}{ccccc} T' \times_T W & \longrightarrow & W & \xrightarrow[b]{} & U \\ \downarrow & & & & \downarrow \\ T' & & \xrightarrow{\quad c \quad} & & X \end{array}$$

is commutative. What this means is that we have proved the existence of a étale locally on T, i.e., we have an $a' : W \to X$. However, since we have proved uniqueness in the first paragraph, we find that this étale local solution satisfies the glueing condition, i.e., we have $\mathrm{pr}_0^* a' = \mathrm{pr}_1^* a'$ as elements of $X(W \times_T W)$. Since X is an étale sheaf we find a unique $a \in X(T)$ restricting to a' on W. □